IRRIGATION WATER MANAGEMENT · NOTE IM01

Surface Irrigation I — Selection of Method, Free Flooding, Border and Check-Basin Irrigation

Course topics: IDE5.5 L16, CAE1.3 L12, IDE-501 T9

Methods of irrigation

A method of irrigation is the way water is applied to the land: by flooding the surface, beneath the surface, by spraying under pressure or in drops.

- **Choose by:** quantity and quality of water, topography, crop, cost of the system and availability of labour.
- **Efficient irrigation** raises yields, maintains soil fertility and saves water. **Over-irrigation** leaches fertiliser, waterlogs the land and builds up salts; too high an application rate, or surface irrigation on sloping land, erodes the soil.
- **The essential requirement:** apply the right amount, at the right time, uniformly, to bring the root zone up to field capacity.

Group	Methods
Surface (gravity)	Free flooding; border (straight, contour); check basin (rectangular, contour, ring); furrow (deep furrow — straight level or graded, contour; corrugations — straight, contour)
Sub-surface	Water table maintained below the surface
Sprinkler (pressurised)	Rotating head; perforated pipe
Drip (pressurised)	Emitter type; micro-tube type

- **Surface irrigation** distributes water by open surface flow from a supply channel at the upper end, flowing over the field by gravity. High efficiency needs a well-built distribution system to control the supply and proper land preparation for uniform distribution.
- **Pressurised irrigation** carries water in pipes under pressure and sprinkles it over the land or canopy, or applies it at points, usually as drops — sprinkler and drip irrigation.

Free (wild) flooding

Water is let out of field channels with no levees to guide it, usually across the field at right angles to the natural slope from head ditches at convenient intervals, released in turns for fairly uniform spread. It is ancient and becoming obsolete — the least efficient method, with uneven distribution and heavy water use.

Border irrigation

Divide the field into long strips (**borders**) separated by low ridges. Turn water into the upper end of each strip; it flows down in a thin sheet guided by the ridges. Each strip has little or no cross slope and a uniform gentle slope along its length, an even surface so the water flows at nearly uniform depth, and is irrigated independently.

- **Suits** close-growing crops, some row crops and orchards, and most soils except those with extremely high or low infiltration — where the land can be levelled at reasonable cost without permanently lowering productivity.
- **Advantages:** ridges are cheap to build with simple implements; one man can irrigate a large area; uniform distribution and high efficiency are possible; large streams can be used efficiently; simple to operate; well-laid borders drain the surface well; machines and combines work easily.
- **Disadvantages:** needs precise land levelling and relatively large streams.
- **Straight (down-the-slope) borders** follow the general slope — easier to build and run where the land can be levelled economically. **Contour borders** run across the slope where it exceeds safe limits or the land is undulating: each strip is level crosswise with a uniform longitudinal grade, forming a series of steps.

Hydraulics

- Flow in a border is spatially varied, unsteady open-channel flow; its discharge falls downstream as water infiltrates. Three phases matter: **advance** of the water front, **ponding and storage**, and **recession** of the tail water.
- **Advance:** a decreasing share flows on while the rest soaks in. The rate depends on stream size, infiltration, surface roughness, vegetative retardance, slope and elapsed time — in an existing field, the stream size is the variable you control.
- **Ponding:** the tail may be closed (preferred for small plots) or open; when the front reaches the closed end, water ponds. Get uniform ponding by cutting off the stream at the right time — before, at or after the front reaches the end.
- **Recession:** after cut-off the water recedes downstream at a rate set by when it disappears from the upper end. Plot advance and recession curves (distance on X, elapsed time on Y) on one graph — when they are parallel, distribution is uniform.
- **Infiltration opportunity time** at a point = time the water recedes from it - time the front reached it.

Design factor	Recommendation
Border width	Generally 2 to 10 m; narrower for small streams
Border length	Sandy and sandy loam 50–100 m; medium loam 100–180 m; clay loam and clay 150–300 m
Safe slope	Sandy loam to sandy 0.25–0.65%; medium loam 0.20–0.40%; clay to clay loam 0.05–0.20%
Stream size	By infiltration rate and border width, in litres per second per metre width — larger on coarse soils, smaller on fine soils

Distribution: carry water along a supply channel at the upper end and divert it into the borders (ridges high enough not to overtop) with portable canvas or metal dams or permanent check-and-turnout structures; let it in through siphon tubes, pipe turnouts, spiles or open turnouts, spaced for uniform distribution.

Check-basin irrigation

Apply water to fairly level plots surrounded by small ridges — for grain and fodder crops and row crops where furrows do not suit.

Type	Features
Rectangular check basins	Small rectangular plots ringed by ridges; nearly level land
Contour check basins	Ridges along contours at a vertical interval of 6 to 12 cm with cross ridges at intervals; sloping land; needs little levelling

- **Hydraulics — four stages:** the entrance stream spreads to cover the full width while advancing; the front advances; the level rises once the stream reaches the far end; the water subsides after cut-off. The variables are those of border irrigation; the differences are the initial spreading and the recession. The spreading equation is not well established, so judge basins by observing the spread and adjust stream size, basin size or both for the desired distribution efficiency.
- **Basin size** depends on stream size, soil and infiltration rate — water must cover the whole basin quickly. From 1 m² for vegetables and intensive cultivation to 1 to 2 hectares or more for wetland rice; small basins on sandy, high-infiltration soils, large ones on clays.
- **Stream size** is not critical if the basin is covered quickly; as infiltration rises, increase the stream or reduce the basin; large streams allow large basins.
- **Distribution:** usually one field channel for every two lines of basins, feeding plots on both sides; admit enough water, cut it off and hold it until it soaks in. Measuring the inflow and timing the stream gives high application and storage efficiency — distribution efficiency is the limit.

> **Quarter-time rule:** the whole basin should be covered in one-fourth of the time needed to infiltrate the net depth. Example: 3 cm depth that takes 40 minutes to infiltrate → cover the basin within 10 minutes.

Limitations: wasteful of labour; uneven land over-waters the lows and under-waters the highs; land under channels and ridges yields less and the ditch-and-levee area is large; in row crops the cross ridges hamper intercultivation, so weeding is manual; ridge-making and watering need much labour.

Square and ring basins (orchards)

- **Square basins** are check basins for orchards — square on flat land, along adjacent contours with short cross ridges on slopes; usually one tree per basin (more if spacing and supply allow). Ridges have rounded tops, 30–35 cm high and 60 cm to 1 m wide at the base; water stands 10–15 cm deep during irrigation. Feed basins straight from the supply channel, or on fairly impermeable land let water overflow basin to basin through drop structures — this saves labour but the upper basins get more water.
- **Ring basins** — circular basins round widely spaced trees, small for young plants and enlarged as they grow, usually one per tree; the whole area is not flooded, so water-use efficiency is high.

Sources: ICAR eCourse 'Irrigation Engineering' (IIT Kharagpur) — surface, sprinkler and drip irrigation lessons; ICAR eCourse 'Micro Irrigation Systems Design' (AgriMoon). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The least efficient surface irrigation method is

(a) border
(b) check basin
(c) free flooding
(d) furrow

Q2. Recommended border length for clay loam and clay soils is

(a) 20–50 m
(b) 50–100 m
(c) 100–180 m
(d) 150–300 m

Q3. Infiltration opportunity time at a point is the difference between

(a) advance and recession times
(b) cut-off and start times
(c) ponding and drainage
(d) two irrigations

Q4. Under the quarter-time rule, a basin needing 40 min to infiltrate the net depth should be covered in

(a) 4 min
(b) 10 min
(c) 20 min
(d) 40 min

Q5. Contour check basins have ridges at a vertical interval of about

(a) 1–2 cm
(b) 6–12 cm
(c) 30–45 cm
(d) 1 m

Q6. Ring basins are best for

(a) rice
(b) wheat
(c) widely spaced trees
(d) pastures

Answers

Q1 — (c) free flooding
Q2 — (d) 150–300 m
Q3 — (a) advance and recession times
Q4 — (b) 10 min
Q5 — (b) 6–12 cm
Q6 — (c) widely spaced trees